

RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



RAJALAKSHMI ENGINEERING COLLEGE

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

(REGULATION 2023)

CS23432 Software Construction

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
LABORATORY RECORD NOTEBOOK

Girivasanth V

Name :

2/AIML/FA

Year / Branch / Section :

2116-231501048

Register No. :

4

Semester :

2024-2025

Academic Year :



RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS)
RAJALAKSHMI NAGAR, THANDALAM – 602 105

BONAFIDE CERTIFICATE

NAME GIRIVASANTH V REGISTER NO. 2116-231501048

ACADEMIC YEAR 2024-25 SEMESTER- IV BRANCH: AIML-B.Tech

This Certification is the Bonafide record of work done by the above student
in the **CS23432- Software Construction** Laboratory during
the year 2024 – 2025.

Signature of Faculty -in – Charge

Submitted for the Practical Examination held on _____

Internal Examiner

External Examiner

INDEX

Name: Girivasanth V Branch: AIML Sec: FA Roll no: 231501048

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EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

AIM:

To establish and evaluate the process of setting up an Azure DevOps environment by creating a new organization through the Azure portal.

INSTALLATION

1. Open a web browser and navigate to:

<https://azure.microsoft.com/en-us/get-started/azure-portal>.

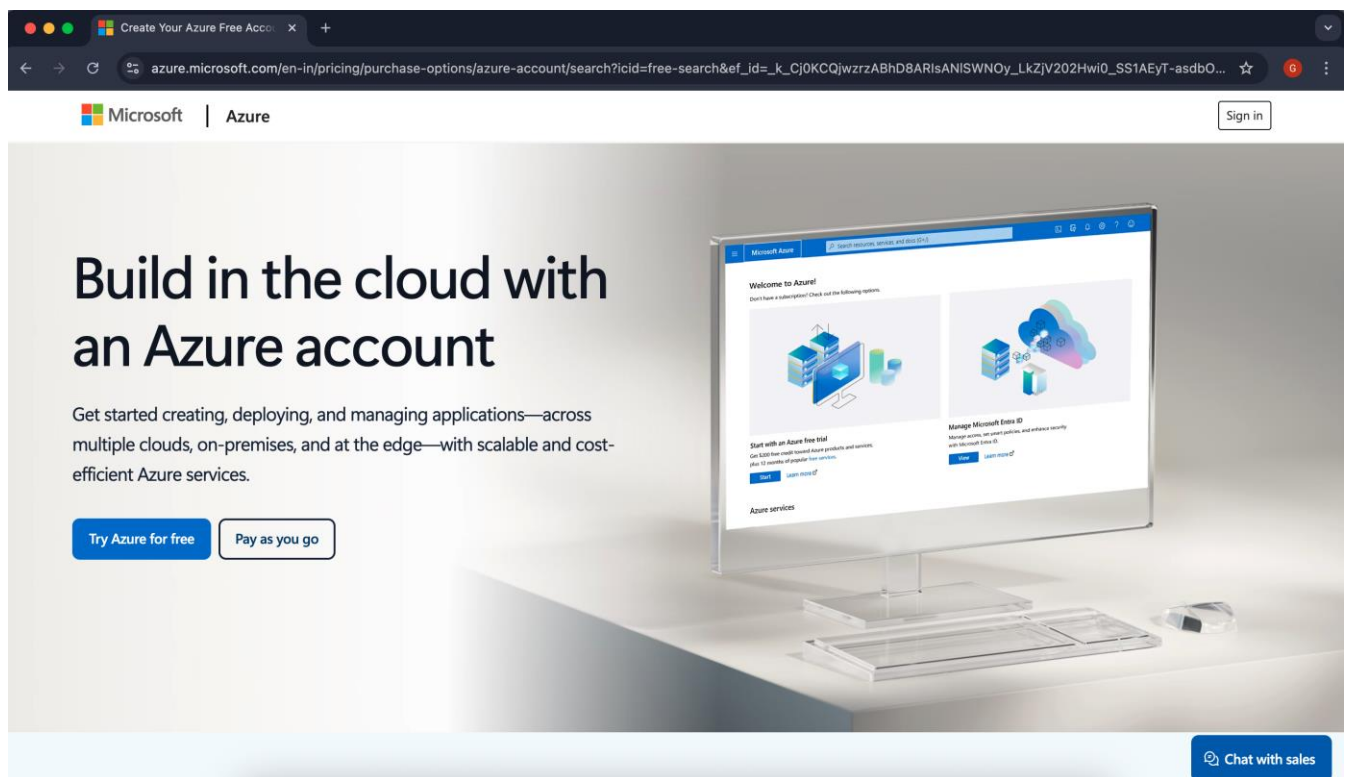
a. **If you have an account:**

Sign in using your Microsoft credentials (e.g., Outlook, Hotmail, or Azure AD account).

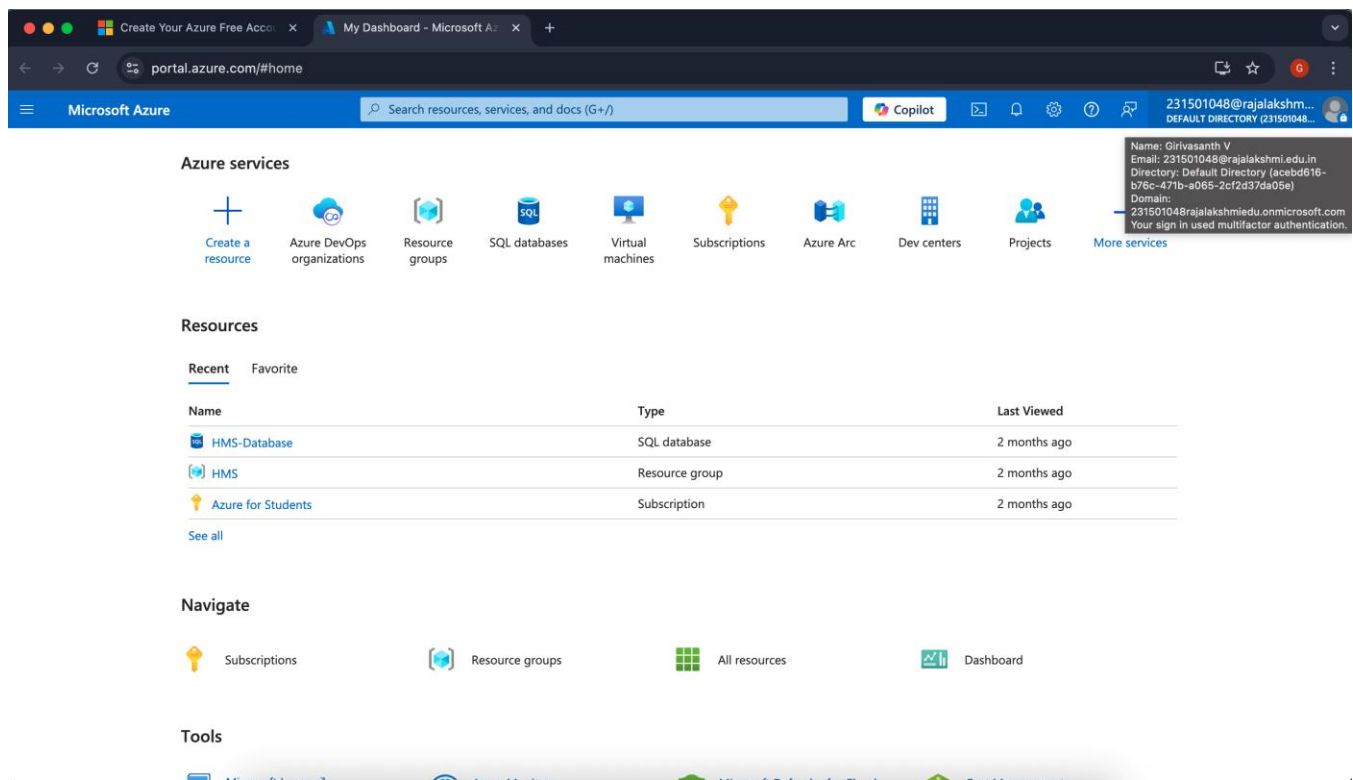
b. **If you don't have an account:**

Create a new Microsoft account at:

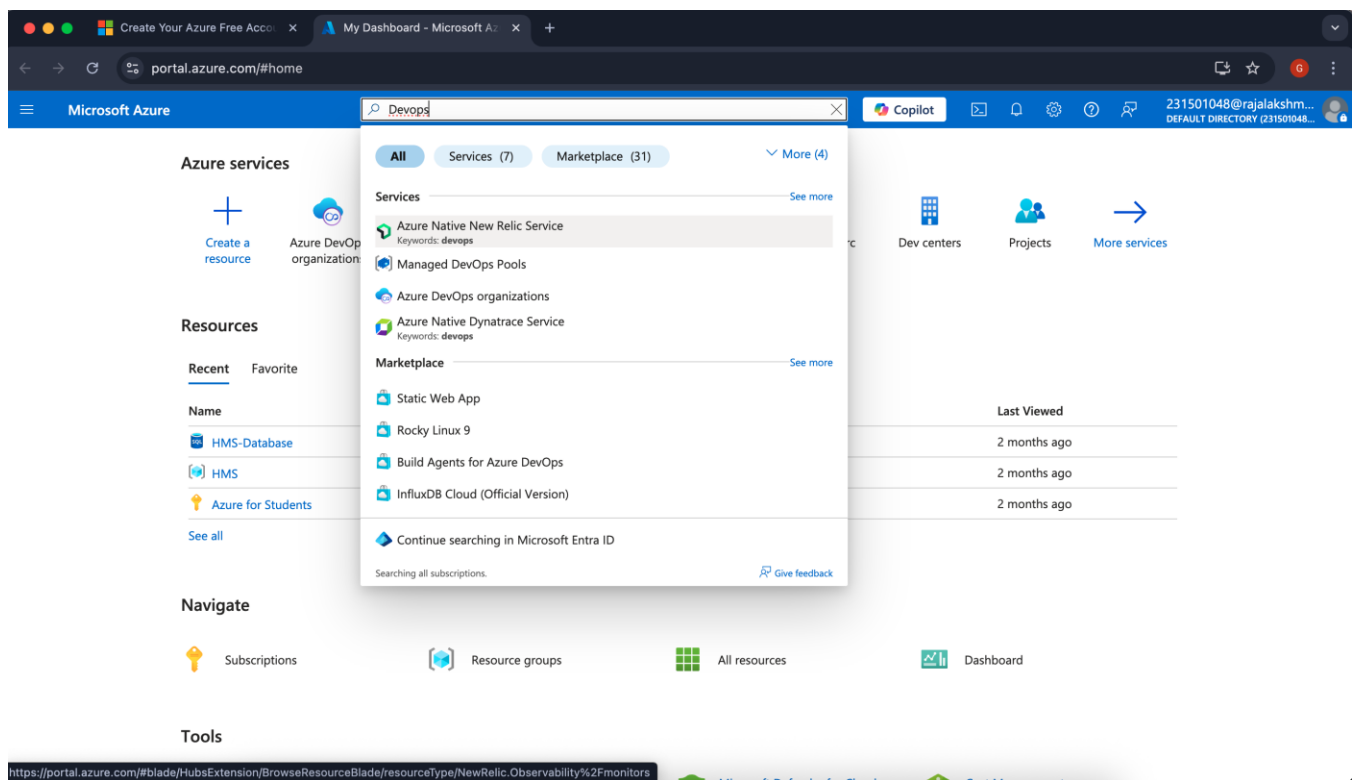
<https://signup.live.com/?lic=1>.



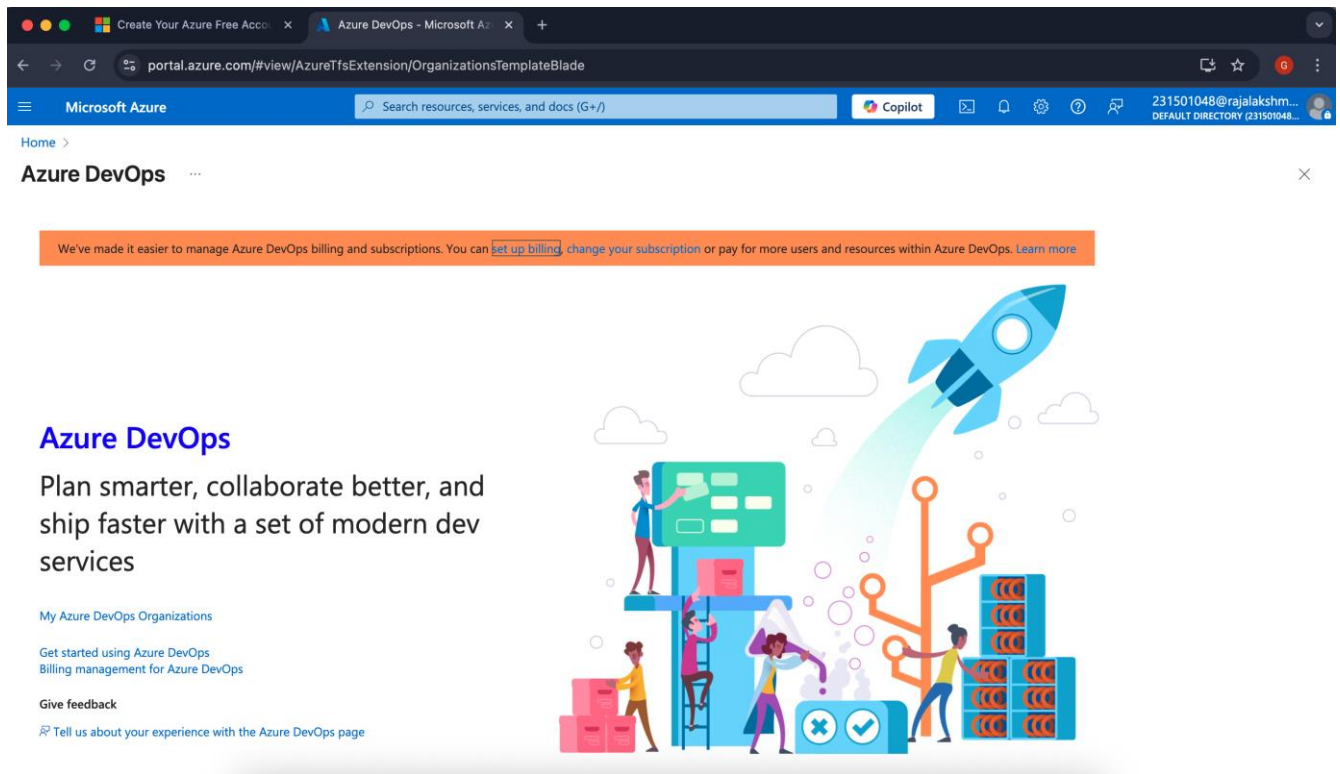
2. Azure home page



3. Search for *Azure DevOps Organizations* in the Azure portal's search bar and select it to open the DevOps environment.



4. Click on 'My Azure DevOps Organizations', set up a new organization, and you'll be automatically directed to the Azure DevOps dashboard.



Result:

The Azure DevOps environment was successfully accessed, and a new organization was created via the Azure portal.

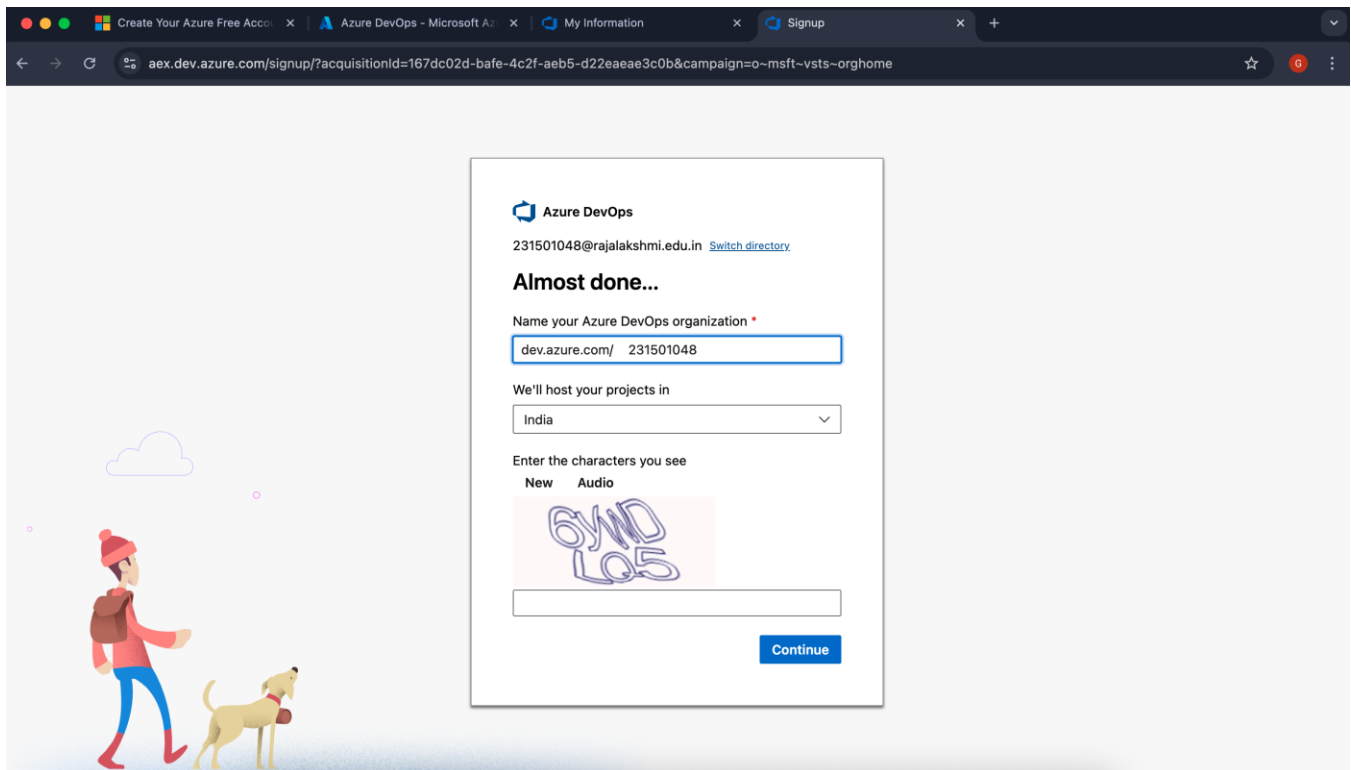
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To establish an Azure DevOps project that enables streamlined collaboration and effective agile project management.

1.Create An Azure Account



2.Create Your First Project in the Organization

a) Initialize Project Setup

Once your organization is established, create your first project to manage code repositories, pipelines, work items, and other development components.

b) Access Project Creation

From your organization's homepage, select "**New Project**" to begin the setup process.

c) Configure Project Details

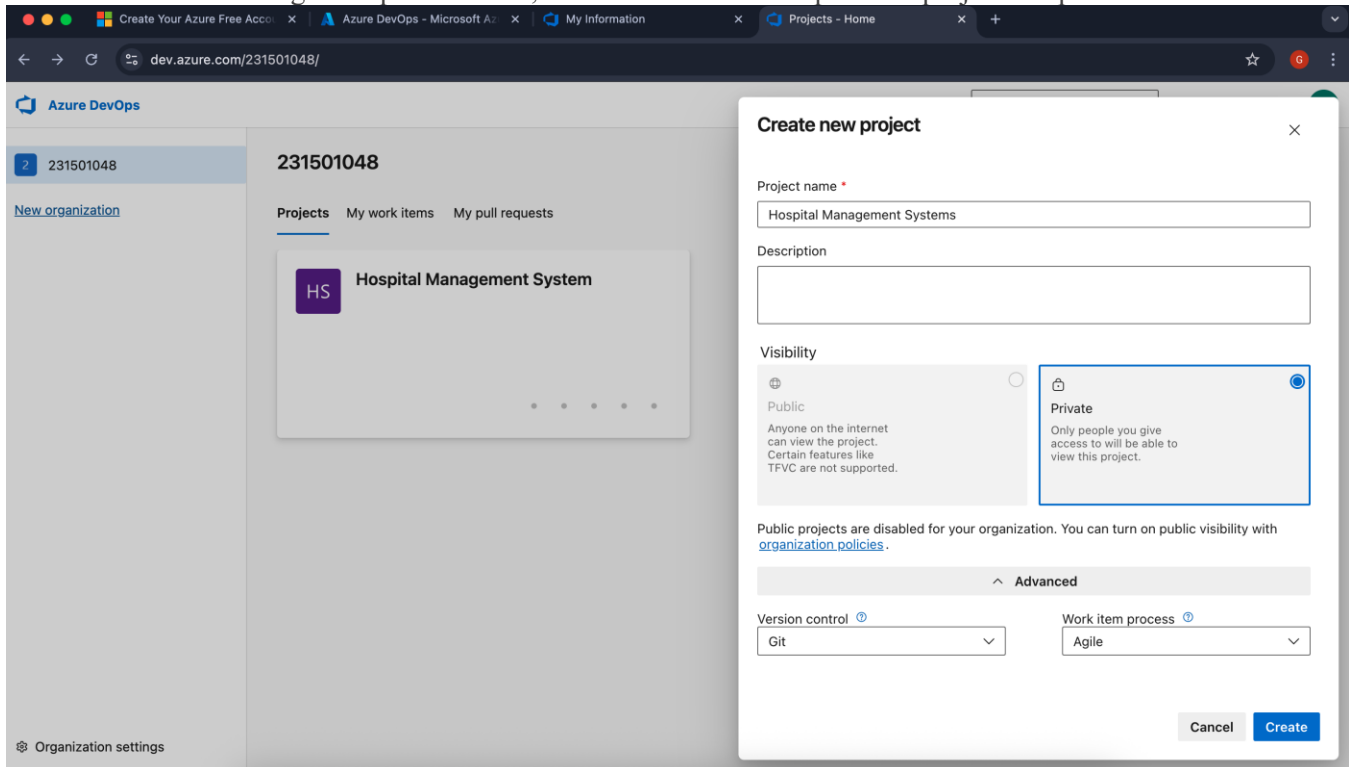
Provide the following information:

- **Name:** Assign a descriptive project name (e.g., "LMS").

- **Description** (Optional): Add relevant details about the project's purpose.
- **Visibility**: Select either:
 - *Private* (Restricted to invited members)
 - *Public* (Visible to all users)

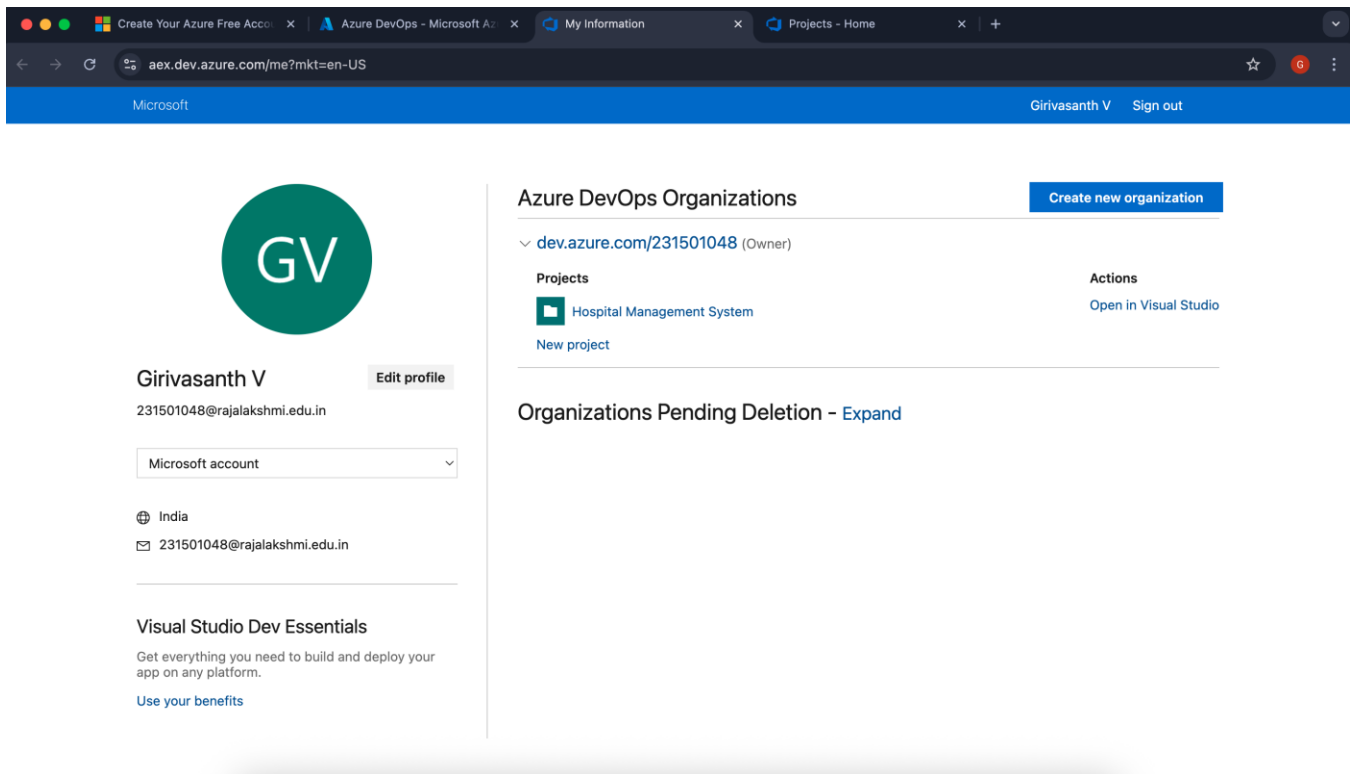
d) d. Finalize Creation

After entering all required details, click "Create" to complete the project setup.

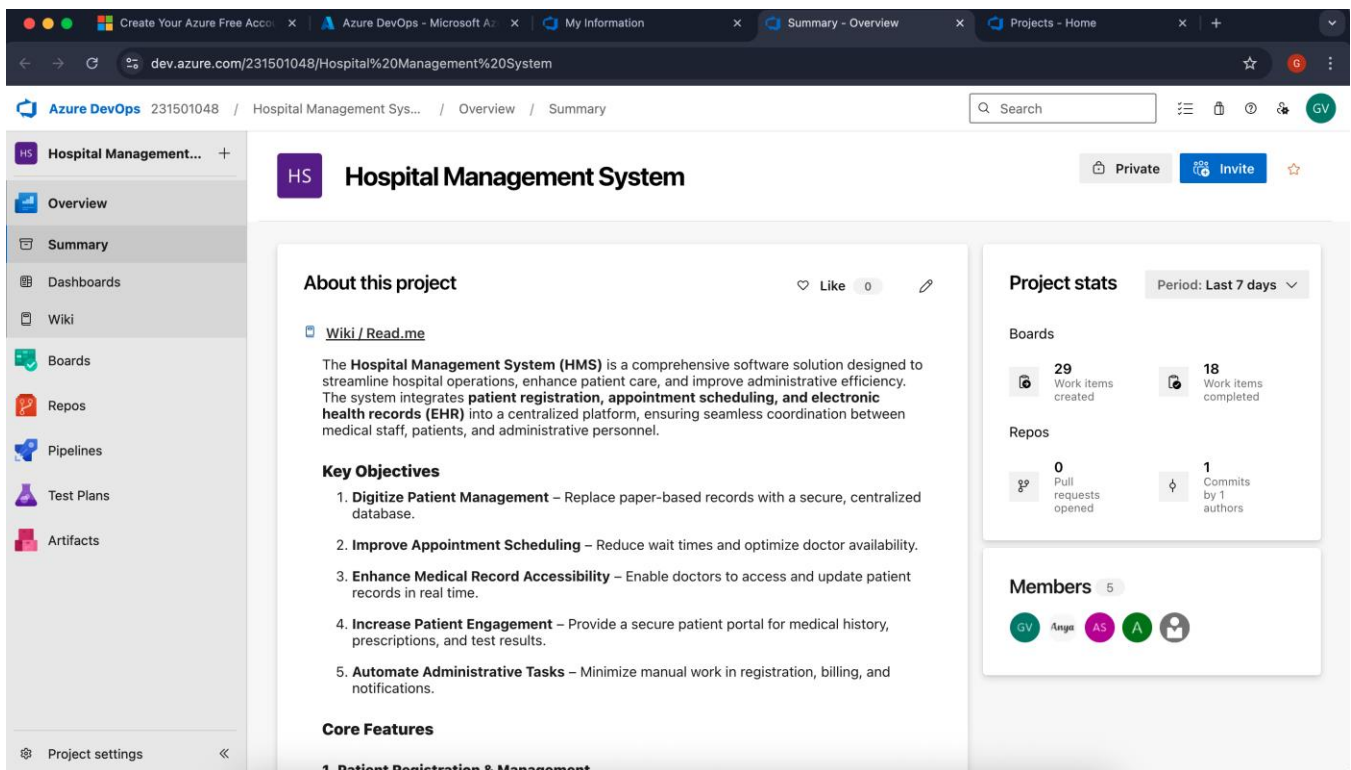


3. After successful login:

- Verify the currently active organization displayed in the top navigation bar
- To change organizations (if you have access to multiple):
 - Locate the organization selector dropdown in the top-left corner (adjacent to your profile icon)
 - Choose your desired organization from the available options
- The system will automatically redirect you to the selected organization's dashboard



4. Project dashboard



5. Creating and Managing User Stories

a. Access Work Boards

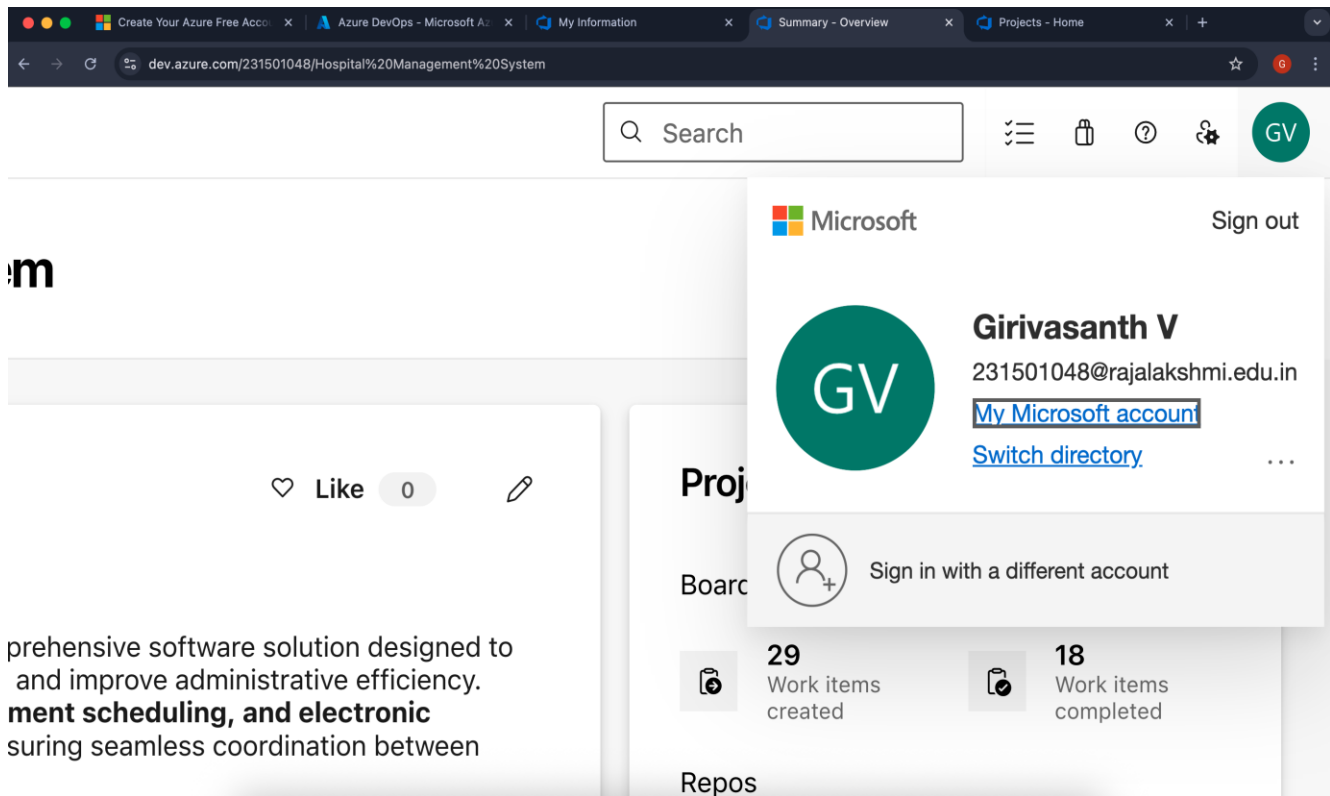
- Navigate to the **Boards** section using the left-hand menu
- This displays your project's work management interface including backlogs, sprints, and work items

b. Create a New User Story

- Locate the **"Add Work Item"** control (typically at the top of the page)
- This may appear as:
 - A "+" button
 - An "Add New Work Item" option
 - A dropdown menu
- Select **"User Story"** from the available work item types
- Complete the form with your story details

The screenshot shows the Azure DevOps interface for a project named 'Hospital Management System'. The left-hand menu is open, showing the 'Boards' section. The main area displays a list of work items under the 'Recently updated' filter. A dropdown menu is open for 'New Work Item', showing options like Bug, Epic, Feature, Issue, Task, Test Case, and User Story. The table lists various work items related to the 'Hospital Management System'.

ID	Title	Assigned To	State	Area Path	Tag
1	Appointment Scheduling System	Girivasanth V	New	Hospital Management System	
2	Appointment Booking	Unassigned	New	Hospital Management System	
3	Appointment Booking by Patients	Adharsh Shankar	New	Hospital Management System	
4	Doctor's Daily Schedule View	Unassigned	New	Hospital Management System	
7	Appointment Reminders	Adharsh Shankar	New	Hospital Management System	
11	Appointment Reminders for Patients	Unassigned	New	Hospital Management System	
12	Electronic Health Records (EHR)	Adharsh Shankar	New	Hospital Management System	
13	Comprehensive Medical Records	Unassigned	New	Hospital Management System	
18	Access and Update Patient Medical Records During Consultations	Adharsh Shankar	New	Hospital Management System	



Result:

The Azure DevOps project has been successfully configured with user story management capabilities and agile workflow implementation.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To create and organize epics, features, user stories, and backlogs in Azure DevOps for streamlined project execution.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a project named 'Hospital Management System Team'. The left sidebar contains navigation options: Overview, Boards, Work Items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, Artifacts, and Project settings. The main area displays the 'Backlog' view with a table of work items:

Order	Work Item Ty...	Title	State	Effort
1	Epic	> Patient Registration Management	New	
2	Epic	> Appointment Scheduling System	New	
3	Epic	> Electronic Health Records (EHR)	New	

On the right, there is a 'Planning' sidebar with a 'New Sprint' button and a summary for 'Sprint 3' (4/21/2025 - 4/28/2025) with a planned effort of 13.6 working days.

1. Create Epic and Fill the Details

The screenshot shows the 'New Epic' form in Azure DevOps. The form has a title field with an error message: 'Field *Title* cannot be empty.' Below the title field is a 'No one selected' dropdown for assignees. The form is divided into several sections:

- Description:** A text area for adding a description.
- Discussion:** A comment box with a 'switch to Markdown editor' link.
- Planning:** Fields for Priority (2), Risk, Effort, Business Value, Time Criticality, Start Date, and Target Date.
- Deployment:** A section for tracking releases, including a 'Add link' button and a note about linking to Azure Repos.
- Classification:** A section for categorizing the work item.

2. Create Feature and Fill the Details

Azure DevOps 231501048 / Hospital Management Sys... / Boards / Work items

NEW FEATURE * • Field 'Title' cannot be empty.

Enter title

No one selected 0 Comments Add Tag Save

State: New Area: Hospital Management System Reason: New Iteration: Hospital Management System\Sprint 1

Description
Click to add Description.

Discussion
Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request. [switch to Markdown editor](#)

Planning
Priority: 2
Risk
Effort
Business Value
Time Criticality
Start Date
Select a date...
Target Date
Select a date...

Deployment
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development
Add link
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

3. Create User Story and Fill the Details

Azure DevOps 231501048 / Hospital Management Sys... / Boards / Work items

NEW USER STORY * • Field 'Title' cannot be empty.

Enter title

No one selected 0 Comments Add Tag Save

State: New Area: Hospital Management System Reason: New Iteration: Hospital Management System\Sprint 1

Description
Click to add Description.

Acceptance Criteria
Click to add Acceptance Criteria.

Discussion
Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning
Story Points
Priority: 2
Risk

Classification
Value area: Business
Story Points

Deployment
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development
Add link
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

4.Create Task and Fill the Details

Azure DevOps 231501048 / Hospital Management Sys... / Boards / Work items

Work Items Back to Work Items

NEW TASK * Field 'Title' cannot be empty.

Enter title

No one selected 0 Comments Add Tag Save

State: New Reason: New Area: Hospital Management System Iteration: Hospital Management System\Sprint 1

Details

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Implementation

Integrated in Build

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

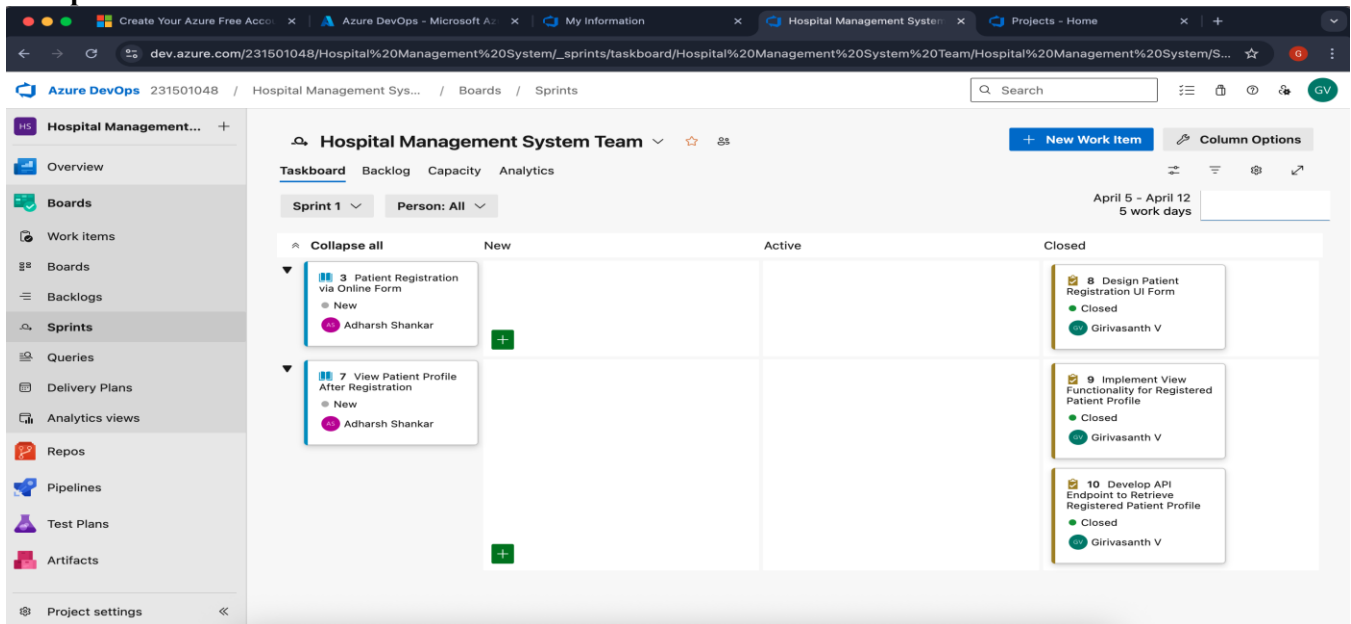
EXP NO: 4

SPRINT PLANNING

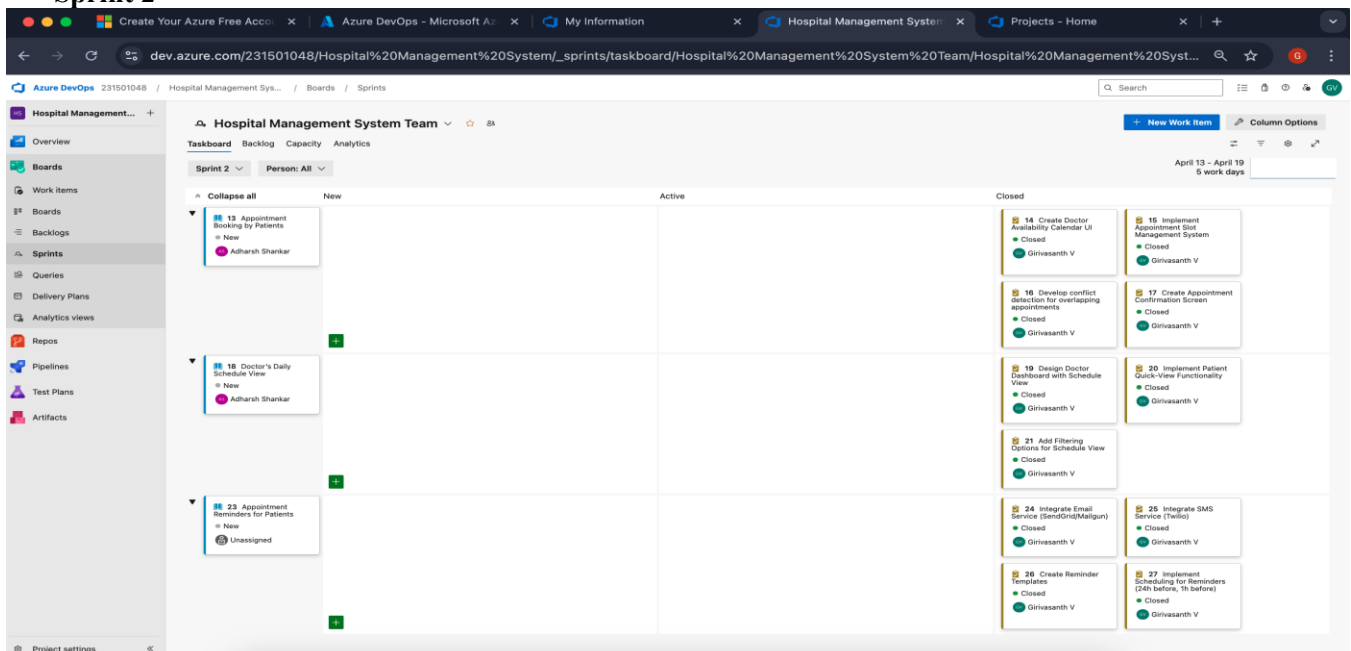
Aim:

To allocate user stories to designated sprints for the Hospital Management System project.

Sprint Planning Sprint 1



Sprint 2



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CS23432

Sprint 3

The screenshot displays the Azure DevOps Sprints interface for the 'Hospital Management System Team'. The browser address bar shows the URL: `dev.azure.com/231501048/Hospital%20Management%20System/_sprints/taskboard/Hospital%20Management%20System%20Team/Hospital%20Management%20System/S...`. The interface includes a left-hand navigation pane with options like Overview, Boards, Work items, Backlogs, Sprints (selected), Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, Artifacts, and Project settings. The main area is titled 'Hospital Management System Team' and shows the 'Sprint 3' board for the period 'April 21 - April 28' with '1 work day remaining'. The board is divided into columns: 'New', 'Active', and 'Closed'. A tooltip for item '30 Access and Update Patient Medical Records During Consultations' is visible, showing it is assigned to 'Adharsh Shankar'. The 'Closed' column contains four items: '31 Design EHR Interface', '38 Implement medical history section', '39 Develop a Prescription Management System for EHR', and '40 Implement Test Results Upload and Viewing System', all marked as 'Closed' and assigned to 'Girivasanth V'.

Hospital Management System Team

Sprint 3 | Person: All | April 21 - April 28 | 1 work day remaining

Columns: New, Active, Closed

Work Items:

- 30 Access and Update Patient Medical Records During Consultations** (New) - Assigned to Adharsh Shankar
- 31 Design EHR Interface** (Closed) - Assigned to Girivasanth V
- 38 Implement medical history section** (Closed) - Assigned to Girivasanth V
- 39 Develop a Prescription Management System for EHR** (Closed) - Assigned to Girivasanth V
- 40 Implement Test Results Upload and Viewing System** (Closed) - Assigned to Girivasanth V

Result:

Agile sprints have been configured for the Hospital Management System initiative.

EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories – Hospital Management System Project.

Poker Estimation

The screenshot displays a web browser window with the URL `dev.azure.com/231501048/Hospital%20Management%20System/_backlogs/backlog/Hospital%20Management%20System%20Team/Epics?workitem=3`. The page title is "USER STORY 3". The story is titled "3 Patient Registration via Online Form" and is created by "Adharsh Shankar". It has 1 comment and an "Add Tag" button. The story is in the "New" state, with the area "Hospital Management System" and iteration "Hospital Management System\Sprint 1". It was updated by "Girivasanth V" 1 hour ago. The story is categorized as "Details" with 3 comments and 0 votes.

Description

As a **new patient**,
I want to **register myself through an online form**,
So that I can be added to the hospital's system and receive a **unique Patient ID** for future medical services

Acceptance Criteria

- 1.The form must collect mandatory fields: Name, Age, Gender, Contact Number, Address.
- 2.Upon successful form submission, a unique Patient ID is generated.
- 3.If the mobile number or email already exists in the database, the system should display a duplicate warning.
- 4.The user receives a confirmation message/email/SMS upon successful registration.
- 5.Uploaded files (if any) must be securely stored and linked to the patient's profile.
- 6.The system must validate formats for email, mobile number, and age (e.g., age should be 0-120).
- 7.Receptionist/Admin should be able to view all patient entries on the dashboard.
- 8.The user should be redirected to a success page showing their Patient ID and next steps.

Planning

Story Points: 8
Priority: 2
Risk:

Classification

Value area: Business
Story Points: 8

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link ▾

Parent

2 New Patient Registration
Updated 12 Apr • New

Child

Result:

Project estimations were determined via Planning Poker, assigning story points to each user story.

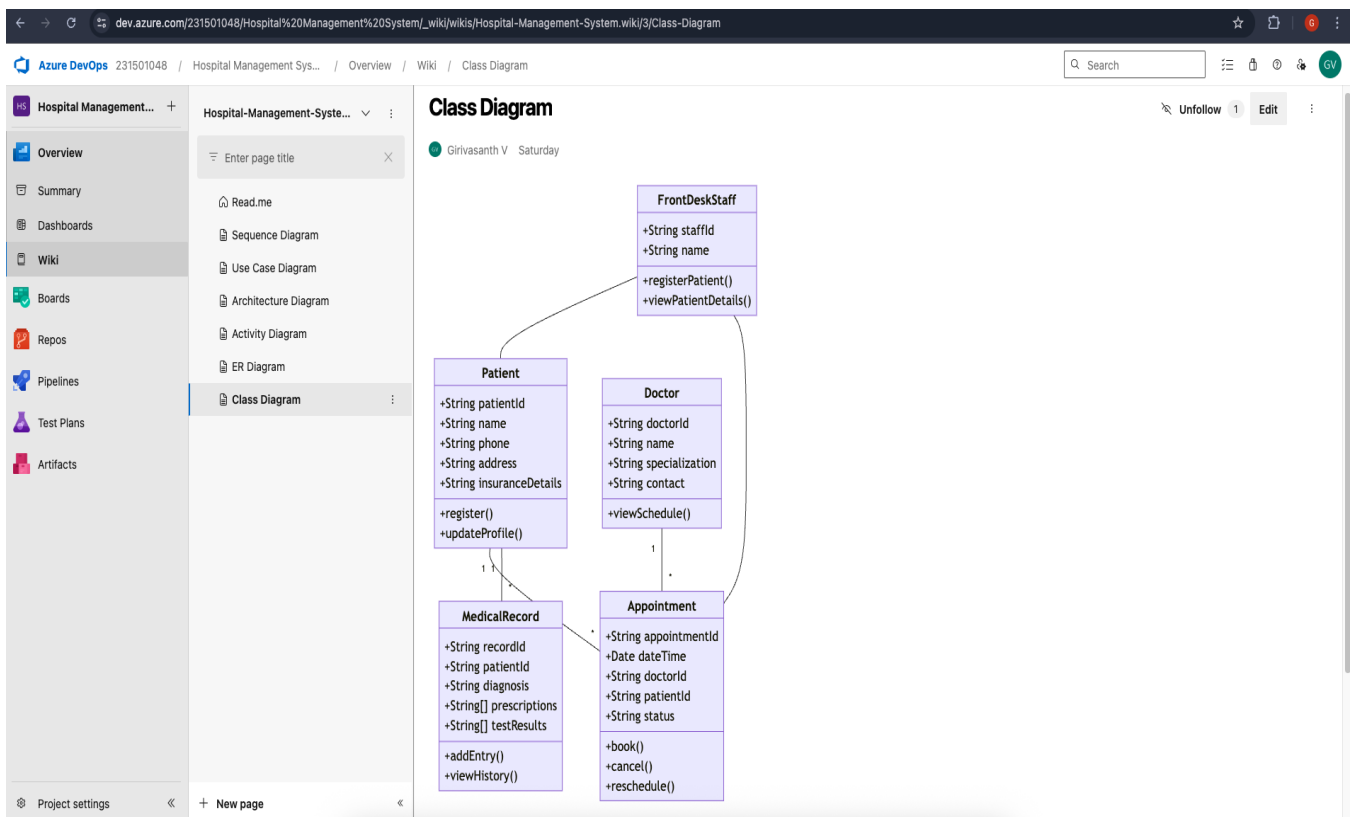
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

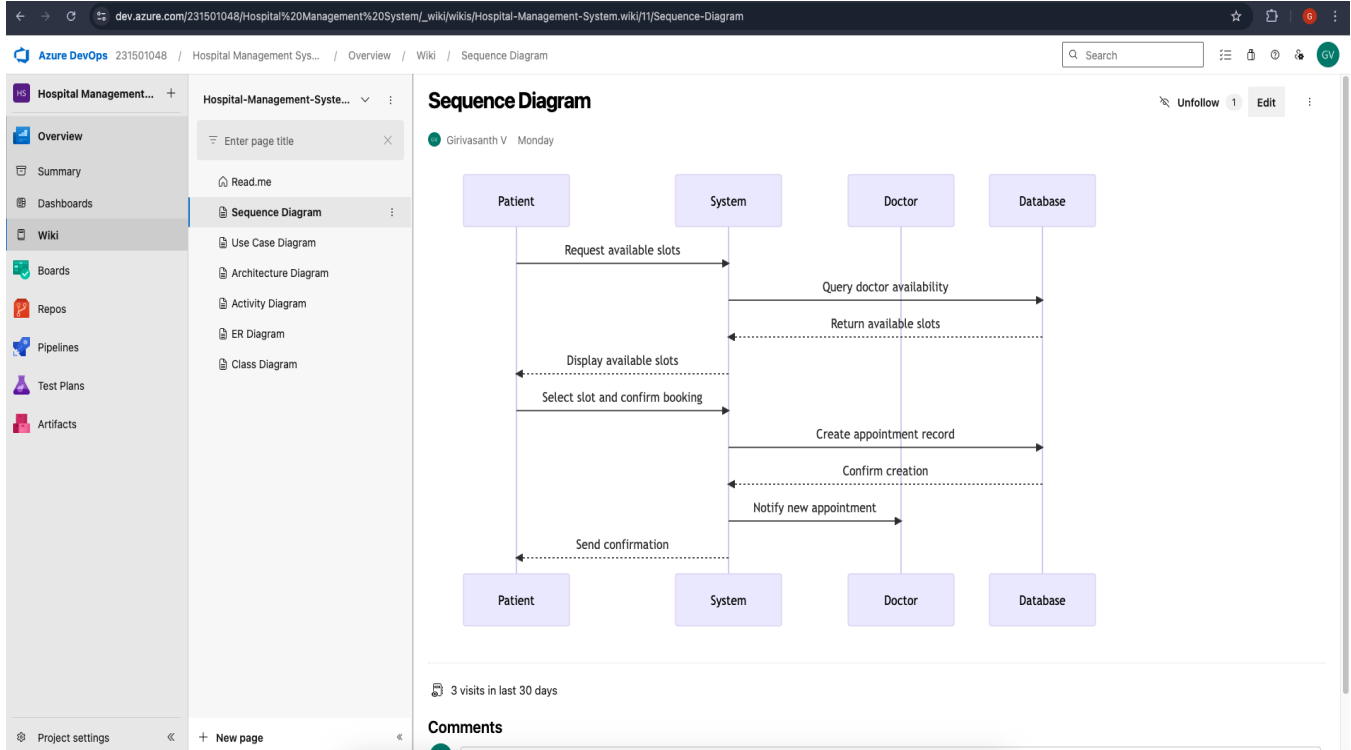
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram have been successfully designed for the Hospital Management System.

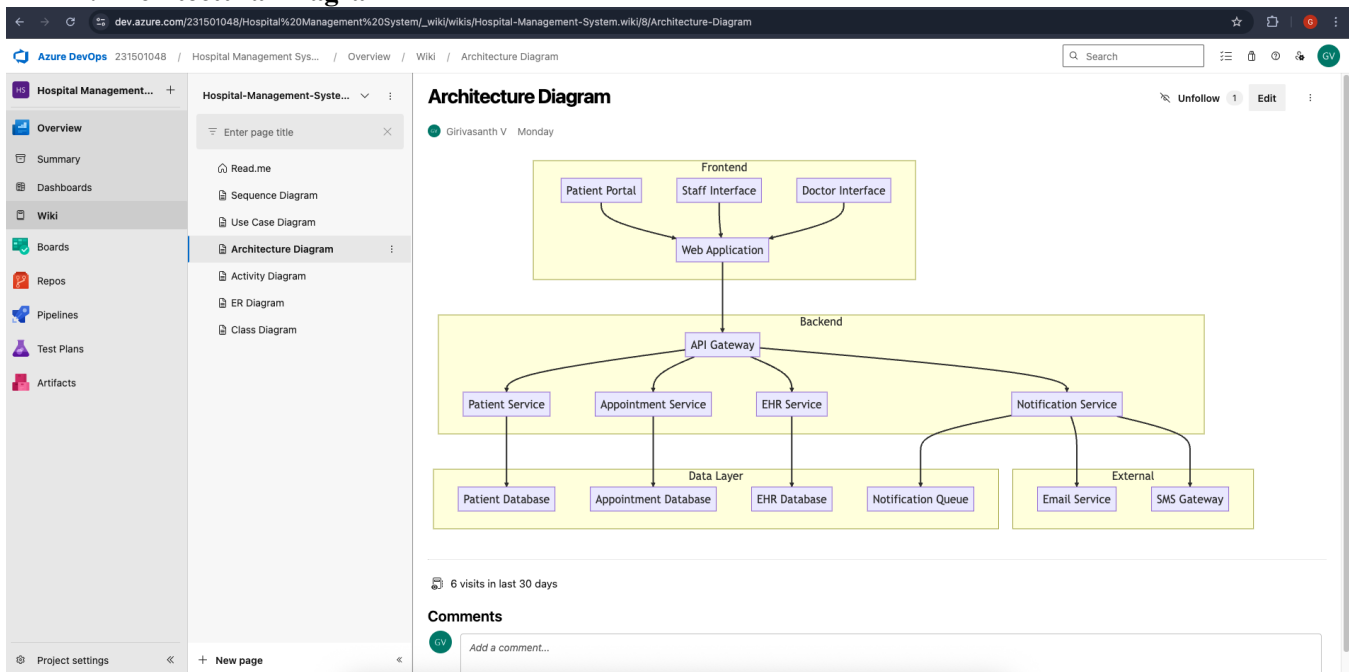
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

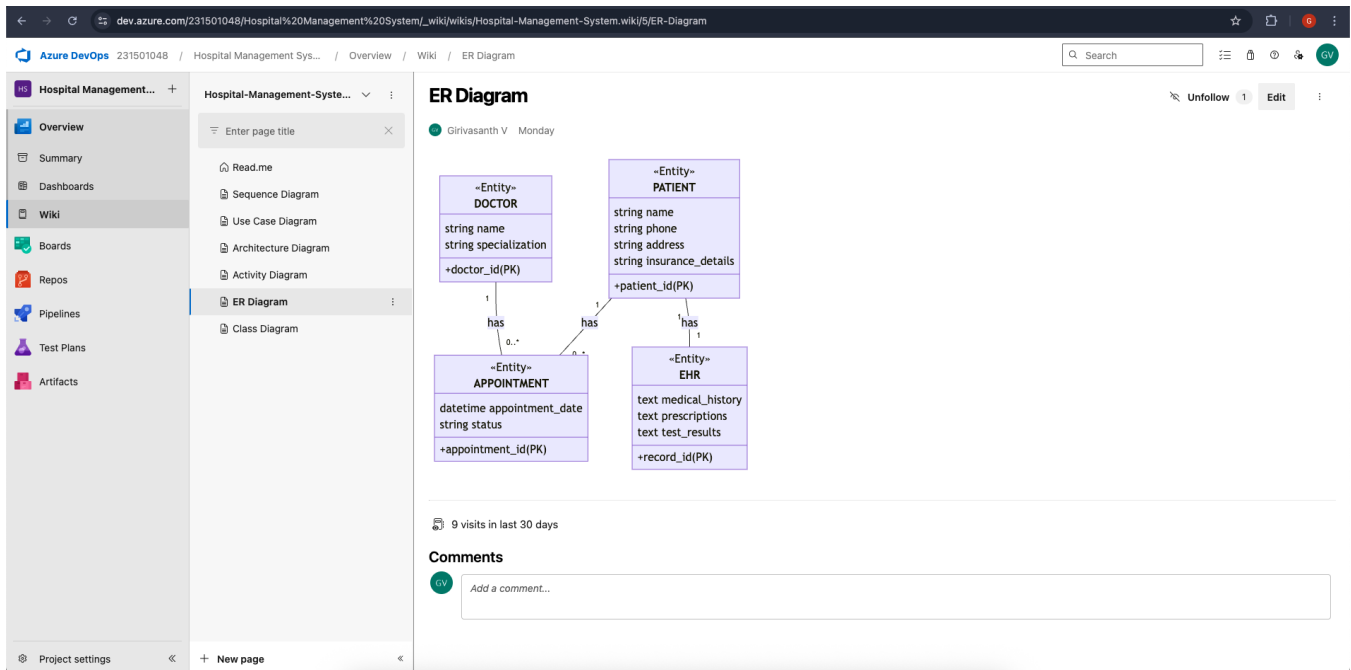
Aim:

Create system architecture and ER diagrams to document the project's design.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram have been successfully designed for the Hospital Management System.

Aim:

Develop test plans in Azure DevOps featuring two test cases per user story (one successful execution path and one error scenario) for five selected user stories.

Test Planning and Test Case:**Test Case Design Procedure****1.Understand Core Features**

- Patient Registration & Management
- Appointment Scheduling
- Electronic Health Records (EHR)
- Doctor Schedule Management
- Reporting & Analytics

2.Define User Interactions

- Each test case simulates real hospital workflows:
 - Front desk registering new patients
 - Patients booking appointments
 - Doctors updating medical records
 - Patients viewing test results

3.Design Happy Path Test Cases

- Validate normal system operations:
 - Successful patient registration
 - Correct appointment booking
 - Proper EHR updates
 - Accurate schedule displays

4.Design Error Path Test Cases

- Test system error handling:
 - Registration with missing fields
 - Booking unavailable time slots
 - Unauthorized EHR access attempts
 - Offline system operations

5.Break Down Steps and Expected Results

- Each test includes:
 - Clear step-by-step actions
 - Specific expected outcomes
 - Example:^[SEP]*Action:* Book appointment *Expected:* Slot marked booked, confirmation sent

6.Use Clear Naming and IDs

- Consistent naming convention:
- TC-REG-01: Successful Registration
- TC-APP-02: Prevent Double-Booking
- TC-EHR-03: Block Unauthorized Access

7.Organize Test Suites

- Group by functionality:
- TS-REG: Registration tests
- TS-APP: Appointment tests
- TS-EHR: Medical records tests
- TS-SEC: Security tests

8. Prioritize and Review

- Mark critical tests (P0):
 - Data security
 - Appointment conflicts
 - EHR accuracy
- Review for:
 - Requirement coverage
 - Edge cases
 - Compliance needs (HIPAA/GDPR)

1. New test plan

The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Runs, and Artifacts. The main area displays a table of test plans under the 'Test Plans' tab. The table has columns for Title, Test Plan ID, State, Area Path, Iteration, and Assigned To. Three test plans are listed, all with a state of 'Active' and assigned to 'Girivasanth V' or 'Adharsh Shankar'.

Title	Test Plan ID	State	Area Path	Iteration	Assigned To
Hospital Management System Team_Stories_Sp...	78	Active	Hospital Management System	Hospital Management System\Sprint 3	Girivasanth V
Hospital Management System Team_Stories_Sp...	66	Active	Hospital Management System	Hospital Management System\Sprint 2	Girivasanth V
Hospital Management System Team_Stories_Sp...	33	Active	Hospital Management System	Hospital Management System\Sprint 1	Adharsh Shankar

2. Test suite

The screenshot shows the Azure DevOps Test Suite execution page. The left sidebar is the same as the previous screenshot. The main area displays the execution results for a test suite titled '23 : Appointment Reminders for Patients (ID: 74)'. The page shows a 'Test Suites' list on the left and a 'Test Points (3 items)' table on the right. The test points table has columns for Title, Outcome, Order, Test Case Id, Configuration, and Tester. All three test points are marked as 'Passed'.

Title	Outcome	Order	Test Case Id	Configuration	Tester
TC11 - SMS Reminder Delivered 24 Hours Before Appointment	Passed	1	73	Windows 10	Girivasanth V
TC12 - Email Reminder Contains Correct Appointment Details	Passed	2	75	Windows 10	Girivasanth V
TC13 - No Reminder Sent for Cancelled Appointments	Passed	3	76	Windows 10	Girivasanth V

3. Test case

Provide two test cases for at least five user stories, demonstrating both happy path (success scenarios) and error path (failure scenarios) in Azure DevOps.

USER STORIES

- As a hospital receptionist or admin I want to view the complete patient profile after registration, So that I can verify the details, make updates if required, and manage the patient's visit efficiently.
- As a **new patient**, I want to **register myself through an online form**, So that I can be added to the hospital's system and receive a **unique Patient ID** for future medical services
- As a **doctor**, I want to **view my daily schedule with patient details** so **I can prepare for consultations**.
- As a patient, I want to receive appointment reminders via SMS or email so I don't miss my appointment.

Test Suites

Test Suite: TS01 - Patient Registration (ID: 91)

1. TC01 – Successful Patient Registration

- **Action:**
 - Go to the Patient Registration page.
 - Enter valid patient details (name, phone, address, insurance).
 - Click "Register".
- **Expected Results:**
 - Registration form is displayed.
 - Fields accept values without error.
 - Unique patient ID is generated, and patient is redirected to confirmation page.
 - **Type:** Happy Path

2. TC02 – Registration with Missing Mandatory Fields

- **Action:**
 - Go to the Patient Registration page.
 - Leave one mandatory field blank.
 - Click "Register".
- **Expected Results:**
 - Error message appears highlighting the missing field.
 - Registration is not completed.
 - **Type:** Error Path

3. TC03 – Duplicate Patient Detection

- **Action:**
 - Register a new patient.
 - Attempt to register the same patient again with identical details.
- **Expected Results:**
 - System detects potential duplicate and shows warning message.
 - Option to proceed or view existing patient record.
 - **Type:** Error Path

Test Suite: TS02 - Appointment Scheduling (ID: 92)

1. TC04 – Successful Appointment Booking

- **Action:**
 - Log in as patient.
 - Select available doctor and time slot.
 - Confirm booking.
- **Expected Results:**

- Appointment is booked successfully.
 - Confirmation appears with appointment details.
 - **Type:** Happy Path
2. **TC05 – Book Already Reserved Time Slot**
- **Action:**
 - Attempt to book an already booked time slot.
 - **Expected Results:**
 - Error message "This time slot is no longer available" appears.
 - System suggests nearest available slots.
 - **Type:** Error Path
3. **TC06 – Appointment Reminder Notification**
- **Action:**
 - Book an appointment for tomorrow.
 - Wait for reminder trigger time.
 - **Expected Results:**
 - SMS/email reminder is sent 24 hours before appointment.
 - Notification contains correct appointment details.
 - **Type:** Happy Path

Test Suite: TS03 - EHR Access (ID: 93)

1. **TC07 – Doctor Access to Patient Records**
- **Action:**
 - Log in as doctor.
 - Search for patient and open EHR.
 - **Expected Results:**
 - Complete patient medical history is displayed.
 - Doctor can add new notes and prescriptions.
 - **Type:** Happy Path
2. **TC08 – Unauthorized EHR Access Attempt**
- **Action:**
 - Log in as non-treating doctor.
 - Attempt to access patient EHR.
 - **Expected Results:**
 - Access is denied with "Unauthorized" message.
 - Attempt is logged in audit trail.
 - **Type:** Error Path
3. **TC09 – Patient Portal EHR View**
- **Action:**
 - Log in as patient.
 - Navigate to "My Health Records".
 - **Expected Results:**
 - Patient can view their complete medical history.
 - No edit options are available.
 - **Type:** Happy Path

Test Suite: TS04 - Doctor Schedule Management (ID: 94)

1. **TC10 – View Daily Schedule**
- **Action:**
 - Log in as doctor.
 - Navigate to "My Schedule".
 - **Expected Results:**
 - All appointments for the day are displayed chronologically.
 - Patient details are visible for each appointment.
 - **Type:** Happy Path
2. **TC11 – Schedule Conflict Detection**

- **Action:**
 - Attempt to schedule overlapping appointments.
- **Expected Results:**
 - System prevents scheduling conflicts.
 - Error message "This would create a schedule conflict" appears.
 - **Type:** Error Path

Test Suite: TS05 - Reporting (ID: 95)

1. TC12 – Generate Patient Report

- **Action:**
 - Log in as admin.
 - Select report parameters (date range, department).
 - Click "Generate Report".
 - **Expected Results:**
 - Report is generated with correct data.
 - Export options (PDF, CSV) are available.
 - **Type:** Happy Path
- #### 2. TC13 – Report Generation with Invalid Parameters
- **Action:**
 - Select future date range for report.
 - Click "Generate Report".
 - **Expected Results:**
 - Error message "Invalid date range selected" appears.
 - Report is not generated.
 - **Type:** Error Path

Test Cases

The screenshot displays the Azure DevOps web interface. The left sidebar shows the navigation menu with options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area shows a work item titled '76 TC13 - No Reminder Sent for Cancelled Appointments' created by 'Girivasanth V'. The work item details include a description, a discussion section with a comment input field, and a deployment section with instructions on how to track releases. The right sidebar shows the 'Steps' tab, 'Summary', 'Associated Automation', and 'Related Work' sections. The bottom of the interface shows the 'Status' and 'Priority' fields.

dev.azure.com/231501048/Hospital%20Management%20System/_workitems/edit/81

Azure DevOps 231501048 / Hospital Management Sys... / Boards / Work Items

Hospital Management... Overview Boards Work Items Boards Backlogs Sprints Queries Delivery Plans Analytics views Repos Pipelines Test Plans Artifacts Project settings

Recently updated Back to Work Items 41 of 47

TEST CASE 81
81 TC15 - Audit Trail Records EHR Modifications

Girivasanth V 0 Comments Add Tag

Save Follow

Updated by Girivasanth V: 43m ago

State Design Area Hospital Management System
Reason New Iteration Hospital Management System(Sprint 3)

Steps

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Add an existing work item as a parent

Tests

30 Access and Update Patient Medical Records During Consultations
Updated 48m ago Active

Status

Priority

4. Installation of test

chromewebstore.google.com/detail/test-feedback/gnldpbnocfnkknnapimkaphdnipb

chrome web store Search extensions and themes

Discover Extensions Themes

Test & Feedback

Featured 4.2★ (175 ratings) Share

Extension Workflow & Planning 200,000 users

Add to Chrome

PCs

Capture & Annotate

change the color of button as per UX needs

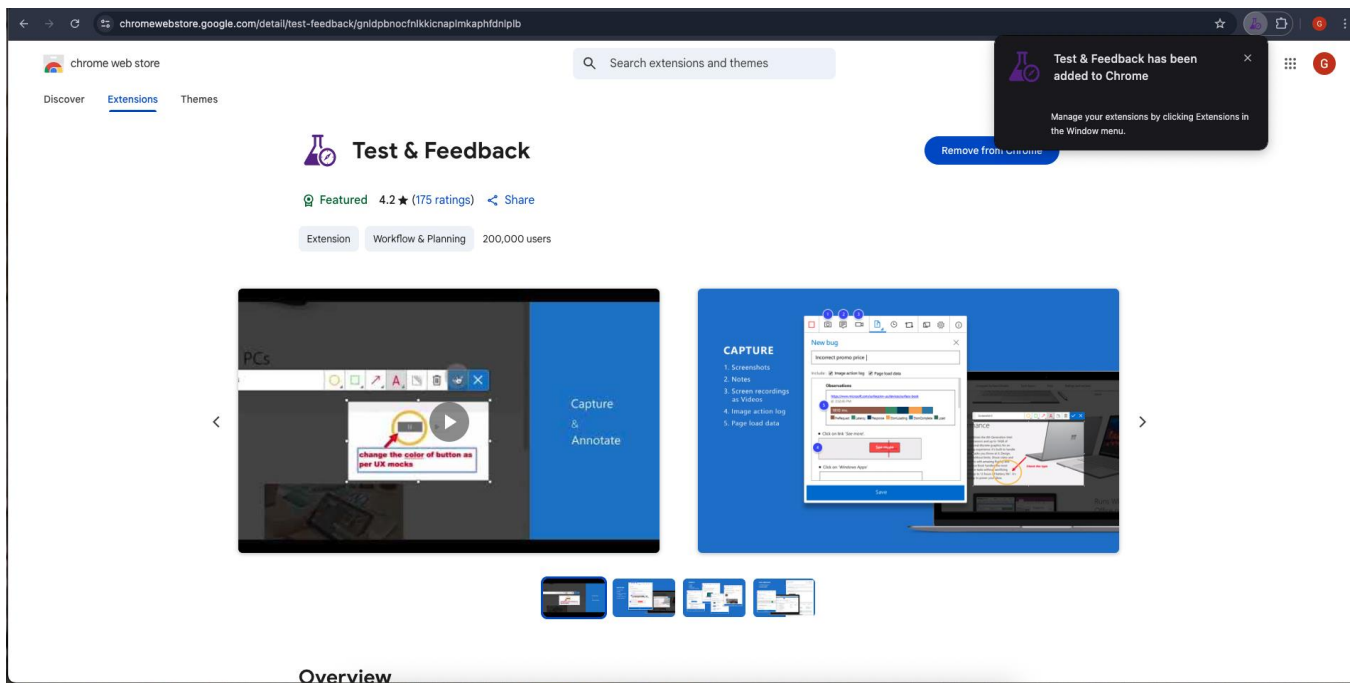
CAPTURE

New bug

Recent private price

1. Screenshots
2. Notes
3. Screen recordings
4. Images as from log
5. Page load data

Overview



Test and feedback Showing it as an extension

Title	Outcome	Order	Test Case Id	Configuration	Tester
TC11 - SMS Reminder Delivered 24 Hours Before Appointment	Passed	1	73	Windows 10	Girivasanth V
TC12 - Email Reminder Contains Correct Appointment Details	Passed	2	75	Windows 10	Girivasanth V
TC13 - No Reminder Sent for Cancelled Appointments	Passed	3	76	Windows 10	Girivasanth V

5. Running the test cases

dev.azure.com/231501048/Hospital%20Management%20System/_testPlans/execute?planid=66&suiteid=74

Azure DevOps 231501048 / Hospital Management Sys... / Test Plans / Hospital Management Sys...

Search

Hospital Management... +

- Overview
- Boards
- Repos
- Pipelines
- Test Plans
- Test plans
- Progress report
- Runs
- Artifacts

Project settings

Hospital Manage...
Apr 13 - Apr 19 **Past**
30% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Hospital Management System Team_Stori...

23 : Appointment Reminders for Patients (ID: 74)

Execute Chart

Test Points (3 items)

Title	Outcome	Order	Test Case Id	Configuration	Tester
☐ TC11 - SMS Reminder Delivered 24 Hours Before Appointment	Passed	1	73	Windows 10	Girivasanth V
☒ TC12 - Email Reminder Contains Correct Appointment Details	Passed	2	75	Windows 10	Girivasanth V
☐ TC13 - No Reminder Sent for Cancelled Appointments	Passed	3	76	Windows 10	Girivasanth V

View execution history

Mark Outcome

Run

Reset test to active

Edit test case

View test result

Run for web application

Run for desktop application

Run with options

Runner - Test Plans

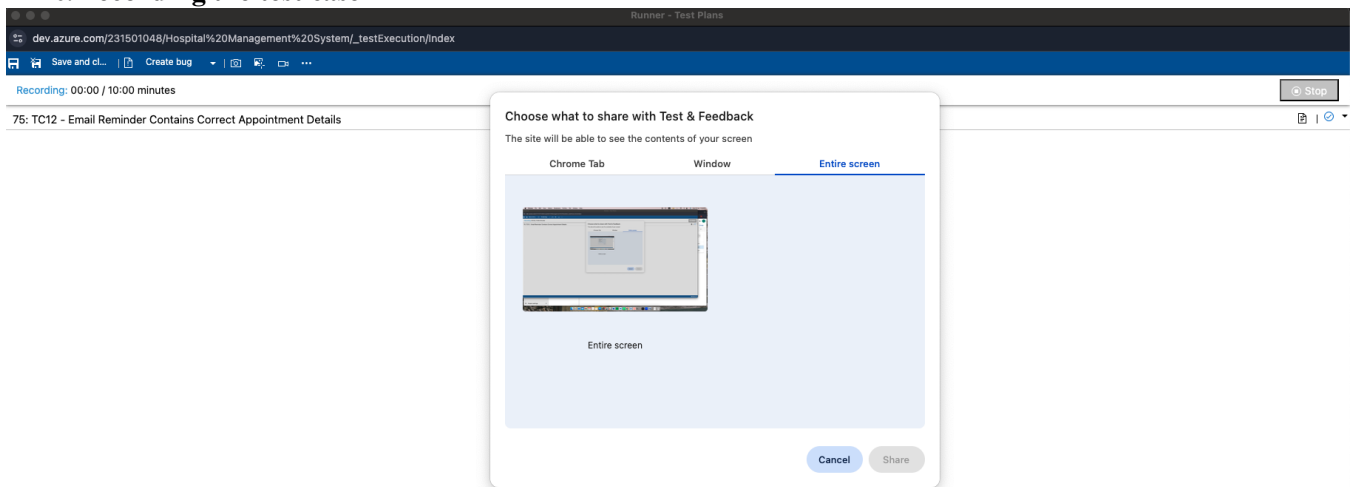
dev.azure.com/231501048/Hospital%20Management%20System/_testExecution/index

Save and cl... Create bug

82: TC16 - Non-Treating Doctor Access Denied to Records

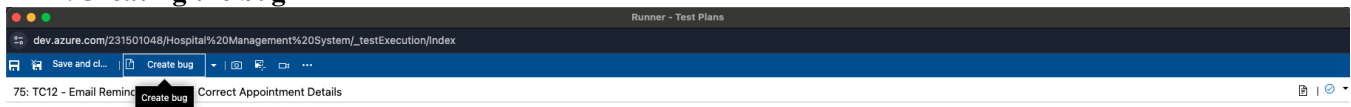
Windows 10

6. Recording the test case

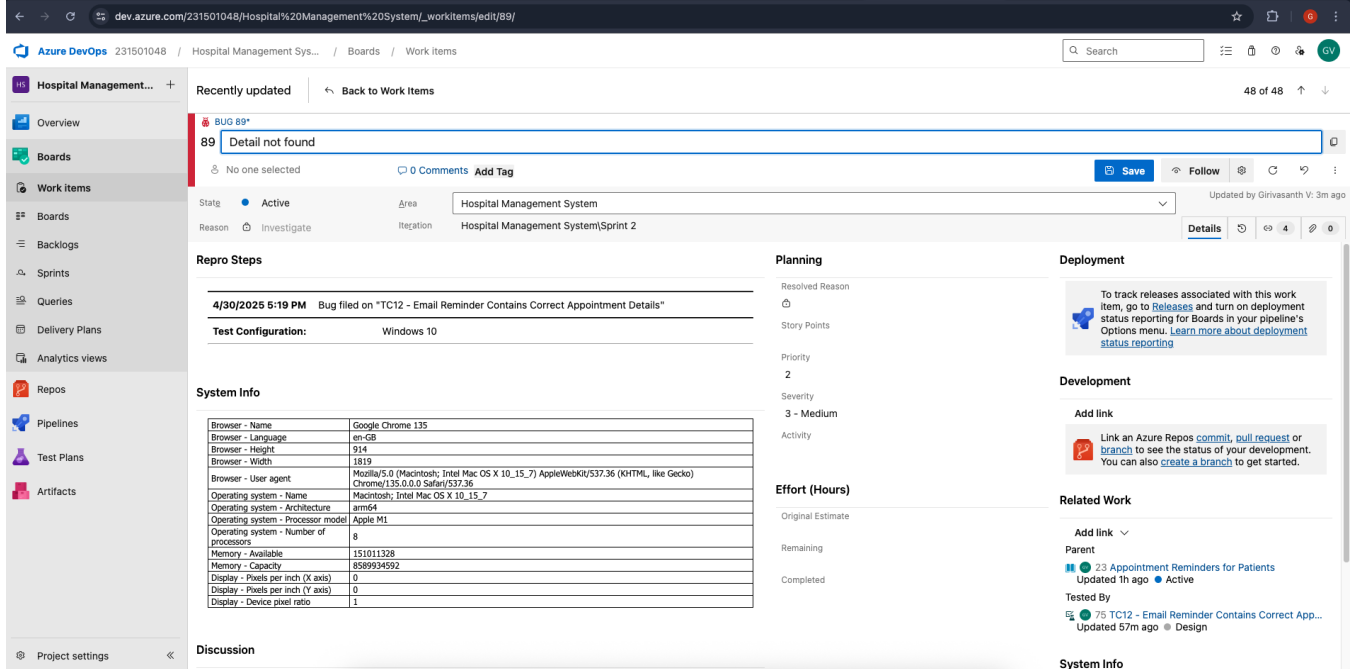


Windows 10

7. Creating the bug



Windows 10



8. Test case results

The screenshot shows the Azure DevOps interface for a test plan. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Runs, and Artifacts. The main area displays the test plan '23 : Appointment Reminders for Patients (ID: 74)' with a status of '30% run, 100% passed'. A modal window titled 'Test Case Results' is open, showing a table of test results.

Outcome	TimeSt...	Configurat...	Run by	Tester	Test
Passed	Just n...	Windows ...	Girivasanth V	Girivasanth V	Hos
Passed	15m ago	Windows ...	Girivasanth V	Girivasanth V	Hos

Below the table, there is a link: [Open execution history for current test point](#).

9. Test report summary

The screenshot shows the Azure DevOps Work Items interface. The left sidebar contains navigation links: Overview, Boards, Work Items, Boards, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays a work item titled '87 Duplicate Patient ID Generation' with a status of 'Active'. The 'Repro Steps' section shows the steps to reproduce the issue. The 'System Info' section provides details about the environment where the bug was reported.

System Info

Browser - Name	Google Chrome 135
Browser - Language	en-GB
Browser - Height	914
Browser - Width	1819
Browser - User agent	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36
Operating system - Name	Macintosh; Intel Mac OS X 10_15_7
Operating system - Architecture	arm64
Operating system - Processor model	Apple M1
Operating system - Number of processors	8
Memory - Available	81838080
Memory - Capacity	8589934592
Display - Pixels per inch (X axis)	0
Display - Pixels per inch (Y axis)	0
Display - Device pixel ratio	1

Discussion

- The bug has been assigned to the developer, and its status has been updated accordingly.

dev.azure.com/231501048/Hospital%20Management%20System/_workitems/edit/89/

Azure DevOps 231501048 / Hospital Management Sys... / Boards / Work Items

Hospital Management... +

Overview

Boards

Work Items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Recently updated

Back to Work Items

48 of 48

BUG 89*

89 Detail not found

Girivasanth V

0 Comments Add Tag

Save Follow

Updated by Girivasanth V: Just now

Details

State: Active

Area: Hospital Management System

Reason: Investigate

Iteration: Hospital Management System\Sprint 2

Repro Steps

4/30/2025 5:19 PM Bug filed on "TC12 - Email Reminder Contains Correct Appointment Details"

Test Configuration: Windows 10

System Info

Browser - Name	Google Chrome 135
Browser - Language	en-GB
Browser - Height	914
Browser - Width	1819
Browser - User agent	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36
Operating system - Name	Macintosh; Intel Mac OS X 10_15_7
Operating system - Architecture	arm64
Operating system - Processor model	Apple M1
Operating system - Number of processors	8
Memory - Available	151011328
Memory - Capacity	858993492
Display - Pixels per inch (X axis)	0
Display - Pixels per inch (Y axis)	0
Display - Device pixel ratio	1

Discussion

Planning

Resolved Reason

Story Points

Priority 2

Severity 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

23 Appointment Reminders for Patients Updated 1h ago Active

Tested By

75 TC12 - Email Reminder Contains Correct App... Updated 53m ago Design

System Info

10. Progress report

dev.azure.com/231501048/Hospital%20Management%20System/_testManagement/analytics/progressreport

Azure DevOps 231501048 / Hospital Management Sys... / Test Plans / Progress report

Hospital Management... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Runs

Artifacts

Project settings

Progress report

Hospital Management System Team_Stories_Sp...

Test Suites

Outcome

Configuration

Tester

Priority

Assigned To

Summary

1 Test plans

5 Test points

5 (5 / 5) Test points run

100% Run

60% (3 / 5) Pass rate

3 Passed

2 Failed

Outcome trend

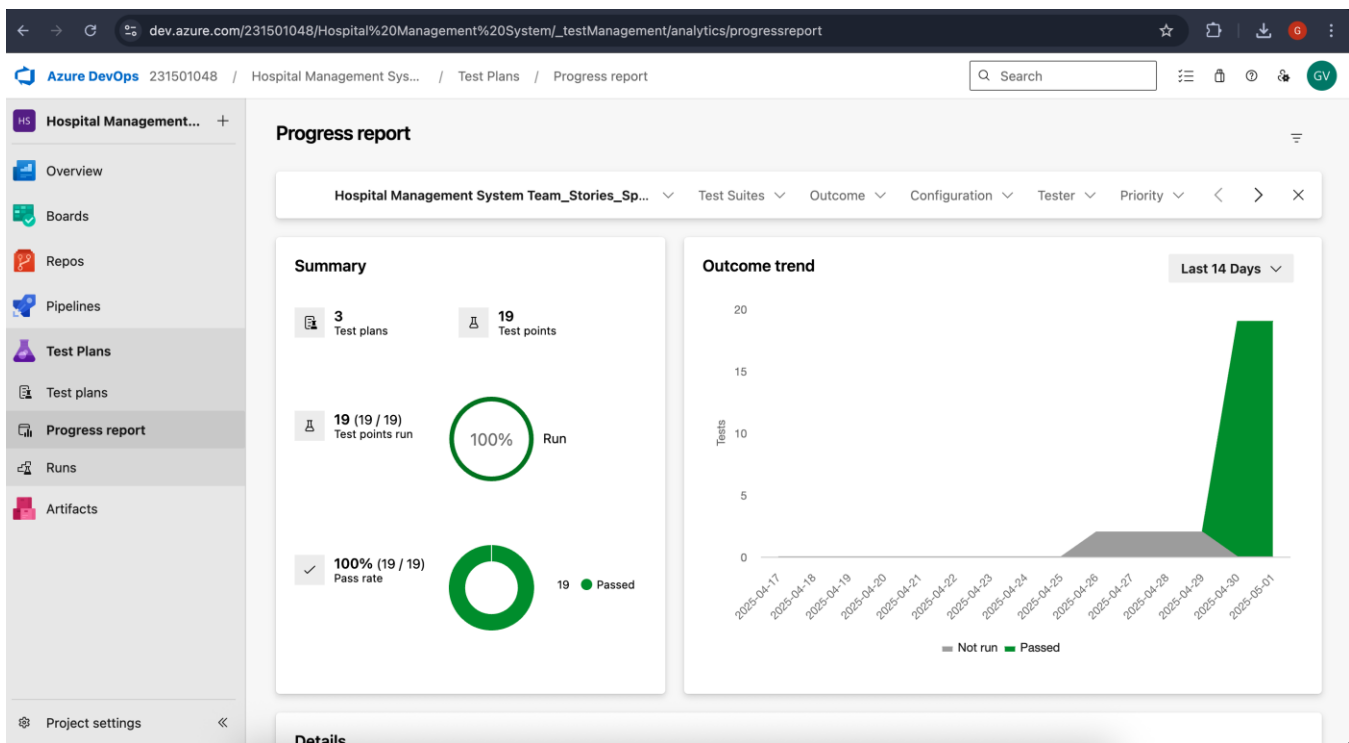
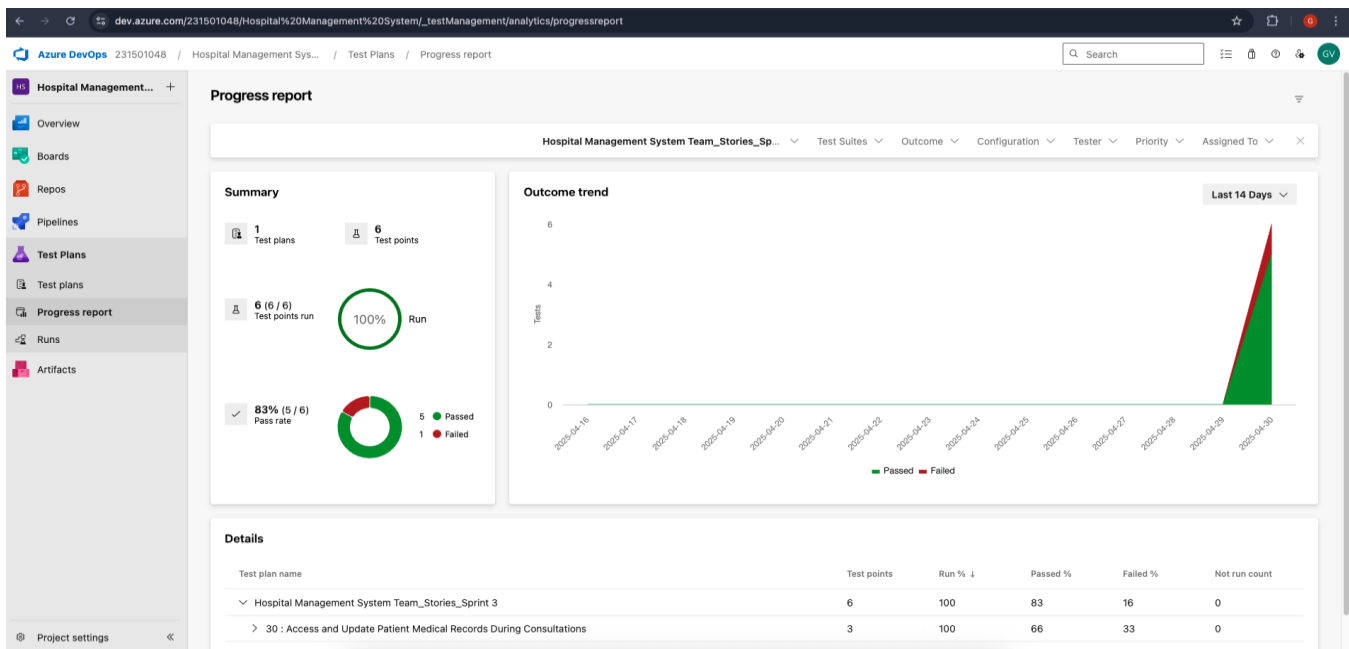
Last 14 Days

Tests

Not run Passed Failed

Details

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
> Hospital Management System Team_Stories_Sprint 1	5	100	60	40	0



Result:

Test plans and test cases covering both Happy Path and Error Path scenarios have been created in Azure DevOps for the respective user stories.

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint.

Load Testing**Steps to Create an Azure Load Testing Resource:****1. Sign in to Azure Portal**

- Navigate to <https://portal.azure.com> and log in.

2. Create the Azure Load Testing Resource

- Go to **Create a resource** → Search for "**Azure Load Testing**".
- Select **Azure Load Testing** and click **Create**.

3. Configure Resource Settings

- **Subscription:** Choose your Azure subscription.
- **Resource Group:** Select an existing one or create a new one.
- **Name:** Enter a unique name (avoid special characters).
- **Location:** Select the preferred region for deployment.

4. (Optional) Add Tags

- Configure tags for better organization and billing tracking.

5. Deploy the Resource

- Click **Review + Create**, then **Create**.
- Once deployment completes, select **Go to resource**.

Steps to Create and Run a Load Test**1. Access Your Load Testing Resource**

- Open your Azure Load Testing resource and select **Add HTTP requests > Create**.

2. Configure Basic Test Settings

- **Test Name:** Provide a unique identifier.
- **Description (Optional):** Briefly describe the test purpose.
- **Run After Creation:** Leave this option checked.

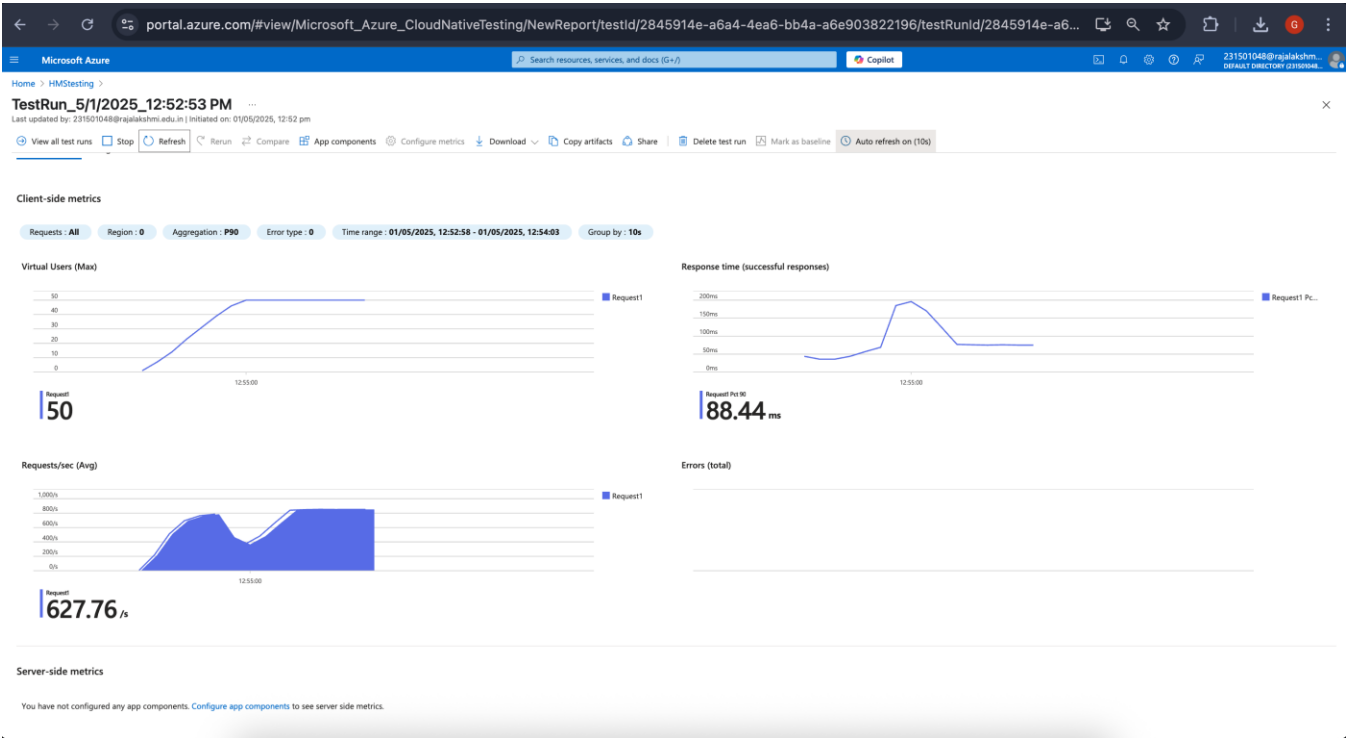
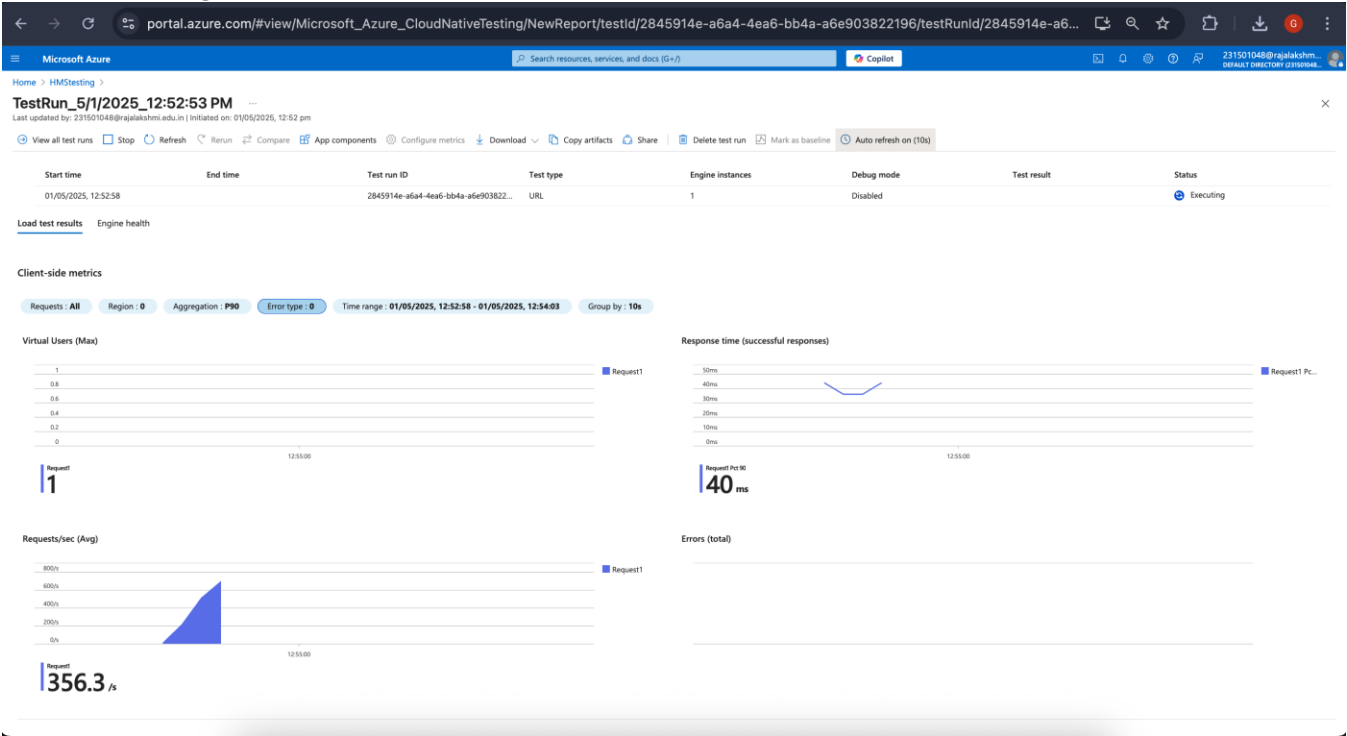
3. Define Load Parameters

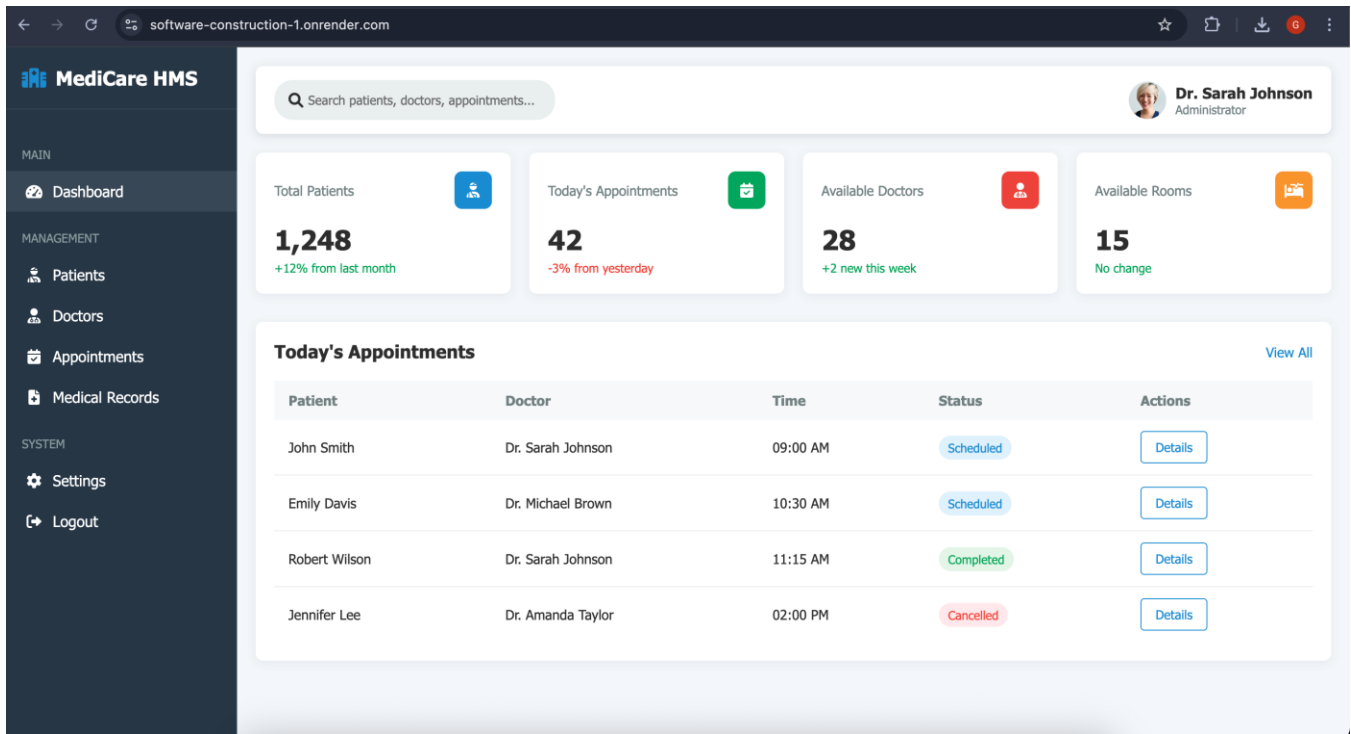
- **Test URL:** Enter the target endpoint (e.g., <https://yourapi.com/products>).

4. Start the Test

- Click **Review + Create** → **Create** to initiate the load test.

Load Testing





Result:

The Azure Load Testing resource is now active, and the performance test for the specified endpoint has been completed.

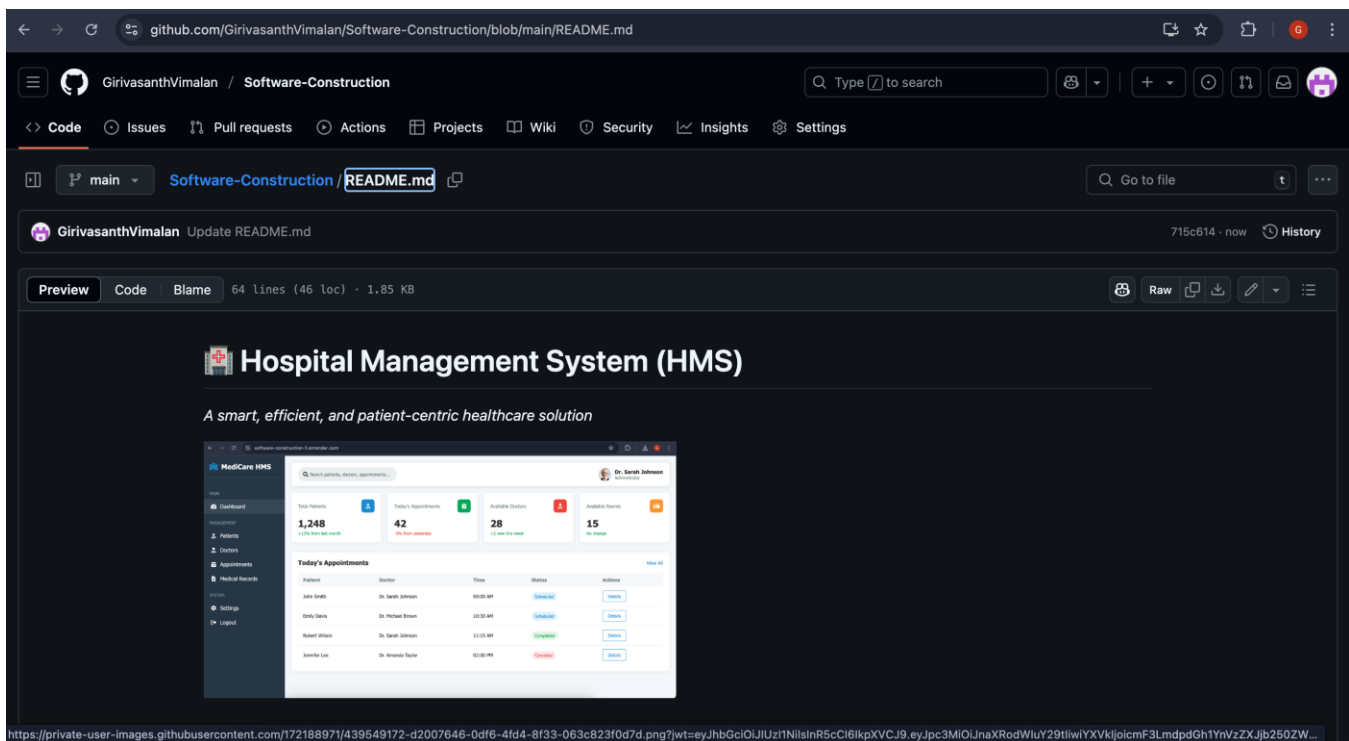
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

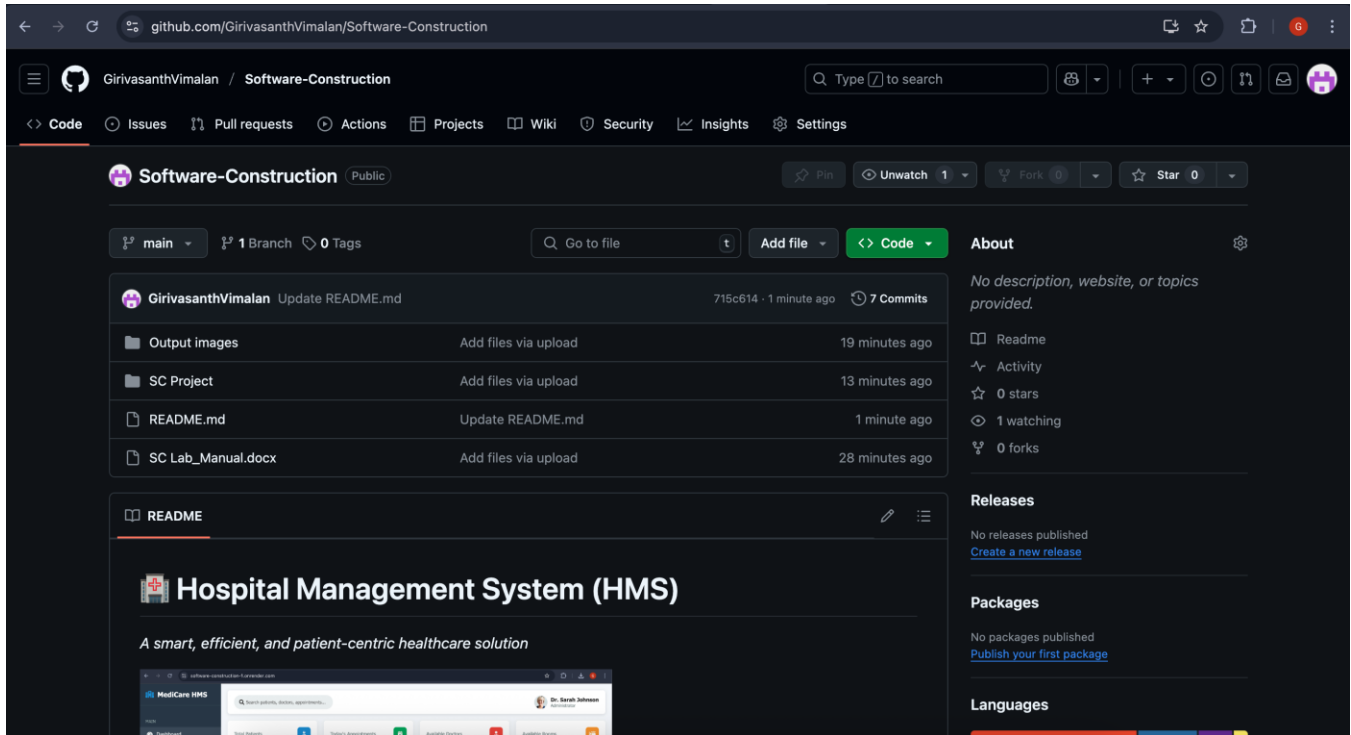
To outline the project's folder structure and file naming conventions in a clear, standardized manner, ensuring that developers, hospital staff, and administrators can easily navigate, maintain, and expand the **Hospital Management System**.

GitHub Project Structure



The screenshot displays the GitHub repository page for 'Software-Construction' by user 'GirivasanthVimalan'. The main content is the README.md file, which features a title 'Hospital Management System (HMS)' and a subtitle 'A smart, efficient, and patient-centric healthcare solution'. Below the text is a screenshot of the HMS dashboard, showing various metrics and a table of appointments.

Patient	Doctor	Time	Status	Address
Jane Smith	Dr. Sarah Johnson	09:00 AM	Completed	123 Main St
Emily Davis	Dr. Michael Brown	10:30 AM	Pending	456 Oak St
Robert Wilson	Dr. Sarah Johnson	11:00 AM	Completed	789 Pine St
Jessica Lee	Dr. Amanda Taylor	02:00 PM	Pending	101 Elm St



Result:

The repository maintains a logical folder structure and standardized naming conventions, creating an intuitive experience for developers navigating the project.